

Edge-AI-Based Adaptive Traffic Signal Control for Emergency Vehicle Priority

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Abstract—Urban traffic congestion can increase vehicle waiting time and create significant delays for emergency vehicles such as ambulances. Conventional traffic signals generally operate using fixed timing schedules and therefore have limited ability to respond to rapidly changing traffic conditions.

This paper proposes an Edge-AI-based adaptive traffic signal control framework that uses roadside cameras and local computing units to analyze traffic conditions and generate traffic-signal decisions. The proposed framework additionally incorporates an emergency vehicle priority mechanism to provide a clearer path through controlled intersections.

The architecture consists of traffic cameras, an edge-processing unit, an emergency vehicle detection module, a decision module, traffic-light controllers, and an optional cloud server for storage and large-scale analysis. The use of local processing can reduce dependence on continuous transmission of raw camera data to a remote server.

The proposed framework provides a foundation for intelligent transportation applications in which traffic signals can adapt to changing traffic conditions while giving priority to detected emergency vehicles.

Index Terms—Edge AI, adaptive traffic signal control, intelligent transportation system, machine vision, emergency vehicle priority, traffic management, computer vision.

I. INTRODUCTION

Traffic congestion is an important challenge in modern urban transportation systems. Conventional traffic lights commonly operate according to predefined timing schedules. Such systems are relatively simple but may not respond effectively when the traffic density changes unexpectedly.

Artificial intelligence and computer vision provide opportunities to develop traffic signal systems that respond to information obtained from the road environment. Camera-based systems can estimate the number of vehicles, queue length, and traffic density, while intelligent algorithms can use these parameters to support adaptive signal decisions.

Research has investigated artificial intelligence and deep reinforcement learning techniques for traffic signal control [1]. Machine-vision-based traffic monitoring and adaptive traffic-light control have also been studied [2].

The handout used as the basis for this work describes a local smart-camera approach in which traffic information is processed near the road and the resulting decision is communicated to the traffic-light controller. It also identifies automatic emergency vehicle path clearance as an important research gap.

Therefore, this paper proposes an Edge-AI-based adaptive traffic signal framework that combines normal traffic adaptation with emergency vehicle priority.

II. LITERATURE REVIEW

A. AI-Based Traffic Signal Control

Artificial intelligence has increasingly been investigated for adaptive traffic signal control. Rasheed *et al.* reviewed deep reinforcement learning approaches for traffic signal control and discussed their potential application in intelligent transportation systems [1].

The basic objective of adaptive traffic control is to use information about current traffic conditions rather than relying only on predefined signal schedules.

B. Machine Vision

Machine vision can be used to obtain traffic information from roadside cameras. Vehicle detection and tracking techniques can provide parameters such as vehicle count and traffic density.

Gorodokin *et al.* investigated optimization of adaptive traffic-light control using machine vision and vehicle-related information [2].

C. Edge Computing

Cloud-based systems can provide significant computational resources. However, continuous transmission of camera data to a remote server can increase communication requirements.

Edge computing moves part of the computation closer to the source of data. In the proposed system, the camera information is processed at a local computing unit before a traffic-signal decision is generated.

III. PROBLEM STATEMENT

Traditional fixed-time traffic signals cannot always respond effectively to sudden changes in traffic density.

Furthermore, a normal adaptive traffic signal may not automatically provide priority to an ambulance travelling through a congested intersection.

The problem addressed in this paper is:

To design an Edge-AI-based adaptive traffic signal framework that analyzes local traffic conditions and provides priority to detected emergency vehicles while maintaining safe traffic signal operation.

IV. RESEARCH GAP

Existing studies have explored adaptive traffic signal control using machine vision and artificial intelligence. However, the integration of local traffic analysis with emergency vehicle priority remains an important practical requirement.

The proposed research therefore combines:

- 1) Camera-based traffic monitoring.

- 2) Edge-based AI processing.
- 3) Adaptive traffic signal control.
- 4) Emergency vehicle detection.
- 5) Emergency vehicle priority.

The proposed framework is intended to address these requirements at the system architecture level.

V. OBJECTIVES

The main objectives of the proposed system are:

- To monitor traffic conditions using roadside cameras.
- To process traffic information locally using edge computing.
- To estimate traffic density and vehicle queues.
- To detect emergency vehicles.
- To provide emergency vehicle priority at intersections.
- To reduce dependence on continuous cloud communication.
- To provide a scalable architecture for intelligent transportation systems.

VI. PROPOSED SYSTEM

The proposed system consists of multiple interconnected components. The major components are:

- 1) Street Camera
- 2) Edge Computing Unit
- 3) Vehicle Detection Module
- 4) Traffic Density Estimation Module
- 5) Emergency Vehicle Detection Module
- 6) Decision Module
- 7) Traffic Signal Controller
- 8) Optional Cloud Server

VII. SYSTEM ARCHITECTURE

The proposed architecture is shown in Fig. 1. The architecture follows the basic sequence:

$$C \rightarrow E \rightarrow A \rightarrow D \rightarrow S \quad (1)$$

where:

- C = Camera input,
- E = Edge processing,
- A = Traffic analysis,
- D = Decision module,
- S = Traffic signal control.

VIII. INPUT PARAMETERS

The system can use multiple traffic parameters. Let the traffic state vector be represented by:

$$X = \begin{bmatrix} V \\ Q \\ D \\ E \end{bmatrix} \quad (2)$$

where:

- V = number of detected vehicles,
- Q = estimated queue length,

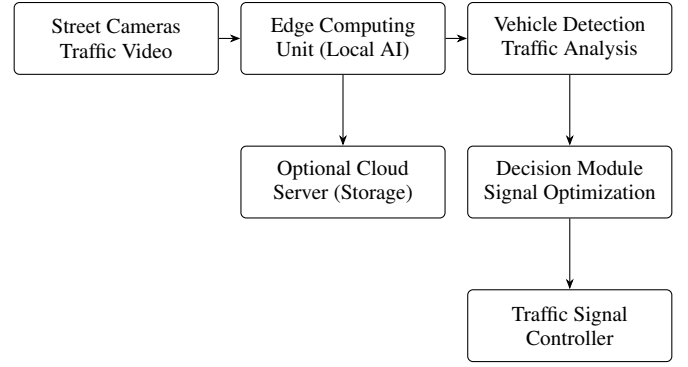


Fig. 1: Proposed Edge-AI-based traffic signal architecture.

- D = traffic density,
- E = emergency vehicle indicator.

For an intersection with multiple lanes, the traffic information can be represented as an array:

$$V = [V_1, V_2, V_3, \dots, V_n] \quad (3)$$

where V_i represents the number of vehicles detected in lane i .

IX. TRAFFIC DENSITY ESTIMATION

Traffic density can be estimated from the number of detected vehicles within a defined road region. A simplified density equation can be written as:

$$D = \frac{N_v}{A_r} \quad (4)$$

where:

- N_v = number of detected vehicles,
- A_r = selected road-region area,
- D = estimated traffic density.

For lane-based analysis:

$$D_i = \frac{V_i}{L_i} \quad (5)$$

where L_i represents the effective road length associated with lane i .

X. EMERGENCY VEHICLE PRIORITY

The emergency vehicle module determines whether an ambulance or another authorized emergency vehicle is present. An emergency indicator can be represented as:

$$E = \begin{cases} 1, & \text{if emergency vehicle is detected} \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

When $E = 1$, the decision module initiates an emergency-priority sequence. The priority condition can be represented as:

$$P = \begin{cases} P_E, & E = 1 \\ P_T, & E = 0 \end{cases} \quad (7)$$

where:

- P_E = emergency priority mode,
- P_T = normal traffic-adaptive mode.

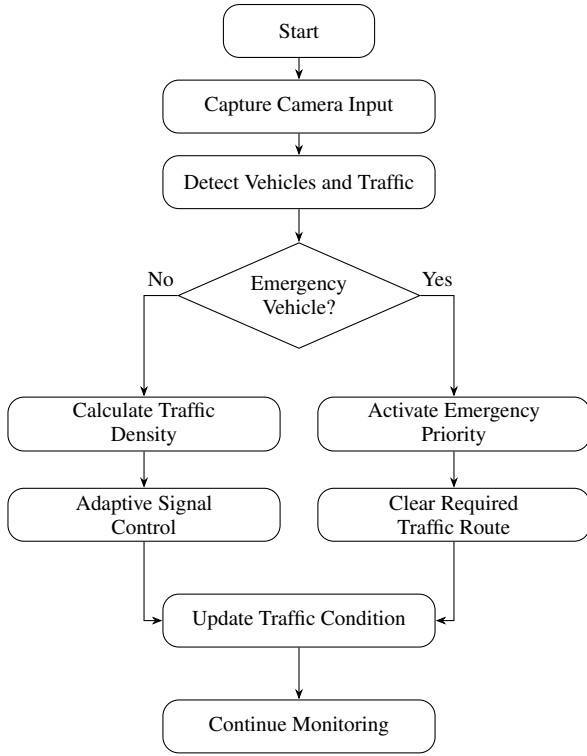


Fig. 2: Flowchart of the proposed adaptive traffic signal system.

XI. SIGNAL DECISION MODEL

The signal decision can consider the traffic density of each approach. Let the traffic-density vector be:

$$D = [D_1, D_2, D_3, D_4] \quad (8)$$

for a four-way intersection. A simplified priority score can be represented as:

$$S_i = w_1 D_i + w_2 Q_i + w_3 E_i \quad (9)$$

where:

- D_i = density on approach i ,
- Q_i = queue length on approach i ,
- E_i = emergency indicator,
- w_1, w_2, w_3 = weighting parameters.

The approach with the highest valid priority score can be considered for signal allocation, subject to traffic safety constraints.

XII. SYSTEM FLOWCHART

The complete working flow is shown in Fig. 2.

XIII. PROPOSED ALGORITHM

The basic operation of the proposed system can be represented using Algorithm 1.

Algorithm 1 Edge-AI Traffic Signal Control

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1: Initialize camera and traffic signal controller
2: Capture traffic image/video
3: Detect vehicles
4: Estimate traffic density and queue length
5: Detect emergency vehicle
6: if emergency vehicle is detected then
7:   Activate emergency priority
8:   Select required traffic direction
9:   Modify signal sequence safely
10: else
11:   Calculate traffic priority
12:   Select appropriate signal phase
13:   Apply adaptive signal timing
14: end if
15: Update traffic information
16: Continue monitoring
  
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TABLE I
COMPARISON OF TRAFFIC SIGNAL APPROACHES

Approach	Decision Method	Internet Dependence	Adaptability
Fixed-Time	Predefined timing	Very Low	Low
Cloud AI	Remote processing	High	High
Edge AI	Local processing	Low/Optional	High

XIV. SYSTEM COMPONENTS

The proposed system can be implemented using a camera, edge-computing platform, traffic signal controller, and appropriate communication interface.

A. Traffic Camera

A roadside camera provides the visual input required for vehicle detection.

XV. TRAFFIC DATA REPRESENTATION USING ARRAYS

Traffic information for multiple lanes can be represented using arrays. For a four-lane intersection, vehicle count can be represented as:

$$V = [25 \quad 42 \quad 18 \quad 31] \quad (10)$$

Similarly, queue length can be represented as:

$$Q = [10 \quad 18 \quad 7 \quad 14] \quad (11)$$

The traffic information can therefore be stored as:

$$T = \begin{bmatrix} V_1 & Q_1 \\ V_2 & Q_2 \\ V_3 & Q_3 \\ V_4 & Q_4 \end{bmatrix} \quad (12)$$

This matrix provides a compact representation of the traffic state at an intersection.

XVI. COMPARISON OF TRAFFIC SIGNAL APPROACHES

Table I compares the major characteristics of different traffic signal approaches.

TABLE II
PARAMETERS USED IN THE PROPOSED SYSTEM

Parameter	Description
Vehicle Count	Number of vehicles detected in a lane
Queue Length	Approximate number of vehicles waiting
Traffic Density	Estimated vehicle concentration
Emergency Indicator	Indicates emergency vehicle detection
Signal Phase	Current traffic-light state
Processing Location	Local edge or cloud processing

XVII. SYSTEM PARAMETERS

The main parameters considered by the proposed system are shown in Table II.

XVIII. PROTOTYPE AND EXPERIMENTAL EVALUATION

A practical implementation of the proposed framework can be developed using a camera, edge-processing computer, AI-based vehicle detection model, and traffic-light controller.

XIX. PERFORMANCE EVALUATION

A future experimental implementation can be evaluated using several measurable parameters.

A. Detection Accuracy

Detection accuracy can be calculated as:

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}} \times 100 \quad (13)$$

where N_{correct} is the number of correctly detected objects and N_{total} is the total number of evaluated objects.

B. Average Waiting Time

Average waiting time can be calculated as:

$$T_{\text{avg}} = \frac{\sum_{i=1}^N T_i}{N} \quad (14)$$

where T_i is the waiting time of vehicle i .

C. Emergency Vehicle Clearance Time

The emergency vehicle clearance time can be represented as:

$$T_c = T_{\text{exit}} - T_{\text{entry}} \quad (15)$$

where:

- T_{entry} = time when the emergency vehicle enters the controlled region,
- T_{exit} = time when the emergency vehicle clears the controlled region.

These equations define evaluation metrics but do not represent measured results unless experimental data are collected.

XX. ADVANTAGES

The proposed framework provides several potential advantages:

- Local processing of traffic information.
- Reduced dependence on continuous cloud communication.
- Adaptive response to changing traffic conditions.
- Emergency vehicle priority.
- Reduced requirement for transmission of raw video data.
- Possibility of multi-intersection coordination.
- Scalable architecture for intelligent transportation systems.

XXI. LIMITATIONS

The proposed system has several limitations.

First, camera-based traffic analysis can be affected by poor lighting, weather conditions, camera positioning, and image quality.

Second, emergency vehicle detection must be sufficiently reliable because incorrect detection may result in unnecessary signal priority.

Third, the traffic signal controller must enforce safety rules before executing AI-generated decisions.

Finally, the numerical performance of the system cannot be established from the proposed architecture alone. Experimental testing is required to determine detection accuracy, response time, traffic waiting-time reduction, and emergency vehicle clearance time.

XXII. FUTURE SCOPE

Future work can include:

- 1) Development of a physical traffic-light prototype.
- 2) Integration of an edge AI computer-vision model.
- 3) Training and testing using traffic datasets.
- 4) Real-time ambulance detection.
- 5) Multi-intersection emergency route coordination.
- 6) Integration with GPS-based emergency vehicle information.
- 7) Development of a traffic monitoring dashboard.
- 8) Testing under different traffic densities and weather conditions.

XXIII. CONCLUSION

This paper presented an Edge-AI-based adaptive traffic signal control framework with emergency vehicle priority. The proposed architecture combines roadside cameras, local AI processing, traffic analysis, emergency vehicle detection, decision-making, and traffic-light control. An optional cloud server can be used for storage and large-scale traffic analysis.

The approach builds on existing research in AI-based traffic signal control and machine vision [1, 2]. The major focus of the proposed system is the integration of adaptive traffic control with emergency vehicle priority.

The mathematical representation, flowchart, algorithm, data arrays, and system architecture provide a basis for implementing and experimentally evaluating the proposed system.

Future experimental work is required to determine actual detection accuracy, signal response time, traffic waiting-time reduction, and emergency vehicle clearance performance.

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